William Gong

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Education

University of Washington Seattle - B.S. Computer Science **September 2023 - June 2027**
Paul G. Allen School - GPA: 3.77/4.0 — Annual Dean's List (2024-2025, 2025-2026) *Seattle, WA*

Tesla STEM High School **September 2019 - June 2023**
High School Diploma - GPA: 4.0/4.0 — SAT: 1570 *Redmond, WA*

Experience

Amazon Web Services (AWS) - EC2 OS **June 2026 - September 2026**
Software Development Engineer Intern *Bellevue, WA*

- Built the entire web interface for the Reporting Center, a new service automating monthly license usage reports for vendored EC2 software (e.g., Windows, Red Hat) – including a **React** frontend and a **GraphQL** layer over the team's **Coral** API – replacing a manual per-vendor email and WorkDocs process.
- Took on backend work for the team after finishing frontend goals early – including extending the Coral API, adding report and notification logging in **DynamoDB**, and fixing bugs across the team's code – achieving a **2x speedup** in report loading.
- Delivered all Reporting Center goals **six weeks ahead of schedule** – including two stakeholder demos, an **MCP** integration (in AppSec review), a fix for a canary timing bug behind multiple Sev-2s, and helping the team adopt AI tools.

ACE-FIRES - ML for Wildfire Spread Forecasting & Sensor Placement **January 2026 - Present**
Research Assistant (Implementation Lead) *UW Disaster Data Science Lab, Seattle, WA*

- Lead implementation for this NSF NCAR-funded project – including helping write the funded proposal and writing a 2,400 GPU-hour NCAR Derecho allocation request to scale training.
- Built a global next-day fire spread dataset of **14,044 fire regions** (2012-2025) – including VIIRS and geostationary satellite fire detections, ERA5/GEFS weather, terrain, and land cover – achieving 80x the coverage of the TS-SatFire benchmark.
- Designed a local-attention transformer for fire spread forecasting in **PyTorch** – including a U-Net stem, attention focused around active fire, and multi-GPU training on Hyak – achieving **test F1 0.40** on unseen 2024-2025 fires.
- Built SENTINEL, a sensor placement system for Jefferson County, CO – including an ignition-risk model, fire and smoke simulation, a budget-constrained optimizer, and a **FastAPI**/**React** map UI – achieving 55% ignition capture in the top 20% of area.

Projects

Fry - Efficient Chess Transformer **September 2026 - Present**
Developer — gilliamwong.github.io/chess-transformer-site *Seattle, WA*

- Designed a **novel transformer architecture** for chess – including SwiGLU MLPs, QK-Norm, a unique form of positional encoding for chess, learned board-relation input features, and a distributional value head – achieving **state-of-the-art training efficiency** and matching leading chess engines in playing ability.
- Benchmarked against Leela Chess Zero's BT4 with MCTS search – including parameter, compute, and dataset comparisons – achieving comparable performance with **8x fewer parameters**, **44x less training compute**, and **1/68th the dataset**.
- Ran **interpretability** experiments on the model's attention – including template ablations, head maps, logit lens, and rollout.

VAR Image Generation with Finite Scalar Quantization **April 2025 - June 2025**
Researcher *Seattle, WA*

- Implemented Finite Scalar Quantization (FSQ) in Visual Autoregressive (VAR) image models in **PyTorch** – including a drop-in replacement for Vector Quantization – achieving stable training at codebook sizes up to 64K entries (vs. the standard 4K limit).
- Trained and evaluated 4 VAR variants on ImageNette with limited compute – including ablations on convergence and codebook utilization – achieving 15-20% lower training loss with FSQ at equal codebook sizes.

Honors & Awards

NSF NCAR Faculty Innovator Program: Contributed to the funded ACE-FIRES proposal ($250K; top 5 of ∼150, 2026)
Wharton Investment Competition: 3rd Place Internationally (2021), Top 50 Internationally as Team Lead (2022)
International Geography Championships: Tied 2nd Place Individual (2022)

Technical Skills

Languages & Tools: Python, Java, C, C++, C#, Rust, SQL, PyTorch, Triton, React, GraphQL, AWS (DynamoDB), NumPy, Slurm
Graduate Coursework: Reinforcement Learning (CSE 542), Intelligent Control Through Learning & Optimization (CSE 579), Social Reinforcement Learning (CSE 599J, in progress)
Relevant Courses: Machine Learning, Deep Learning, Advanced Machine Learning, ML for Neuroscience, Natural Language Processing, Systems Programming, Data Structures & Algorithms, Statistics & Probability, Introduction to Algorithms, Toolkit for Modern Algorithms, Databases, Distributed Systems, Computer Security